

NAYANA NAINENI

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EDUCATION

Bachelor of Science in Data Science: 09/2023 - Present

University of California, San Diego - San Diego, CA

Expected graduation date: May 2027

Relevant Coursework: Data Structures & Algorithms of Data Science, The Practice & Application of Data Science, Systems for Scalable Analysis (Dask, AWS EC2), AI: Probabilistic Models, ML: Learning Algorithms

SKILLS

- **Languages:** Python, Java, HTML/CSS, JavaScript, SQL
- **Frameworks:** Django, React, Flask
- **Libraries:** Pandas, Scikit-learn, Numpy, Librosa, Selenium, LightGBM, Seaborn, Matplotlib, Dask
- **Tools:** Git, IntelliJ, Optuna, Jupyter Notebook, AWS

PERSONAL PROJECTS

Heart Disease Risk Prediction (Python, LightGBM, Scikit-learn, Optuna, Statsmodels)

- Analyzed CDC survey data (N=319K) to predict heart disease risk through statistical and ML methods
- Utilized stratified logistic regression and VIF/Cook's Distance diagnostics to assess risk factors across diabetic and non-diabetic groups
- Tuned LightGBM model with Optuna and stratified K-Fold CV; achieved 76% accuracy and 0.78 macro F1
- Identified 17K+ high-risk undiagnosed individuals for potential early intervention

ReadMatch (Python, Django, SQLite, NumPy, Scikit-learn)

- Developed a book-based matching platform that recommends users using Goodreads data and quiz answers
- Incorporated a cosine similarity algorithm that weighs recently read books heavier for personalization
- Built a Django backend supporting user auth, data import, and REST endpoints for match generation

Family Information Web Application (Python)

- Built a web application to store and manage family member information with Django and SQLite
- Developed responsive interface with HTML/CSS for user-friendly data input and retrieval
- Implemented user authentication, form validation, and search functionality
- Applied OOP principles for clean code and future scalability

Music Recommendation System (Python, Pandas, Scikit-learn, Librosa)

- Created a music recommendation system that analyzes the features of the uploaded song and compares them to a database of other songs through cosine similarity
- Implemented feature extraction techniques using Librosa and recommends the top 5 most similar songs
- Optimized similarity computation for efficient recommendations on larger datasets based on user preferences

Open Source Contributions

- *HuggingFace/LightEval*: Added pytests and integration code for HELMET long-context evaluation task, validating prompt loading and task behavior across JSON dataset files
- *HuggingFace/Transformers*: Improved GGUF tokenizer loading speed for Nemotron test pipelines by switching to NVIDIA tokenizer, reducing CI test runtime
- *HuggingFace/Transformers*: Updated GPT-J model card by improving formatting for documentation

EXTRACURRICULAR ACTIVITIES

Volunteer, 10/2024 - Present

Data Science Student Society

- Contributed to the creation of articles to promote data science knowledge and its real-world applications
- Conducted research on emerging trends in data science, such as machine learning and artificial intelligence
- Collaborated with team members to develop engaging content and encourage active participation within the data science community